

Volume III · Chapter 16 — Deployment, FHIR Integration & Compliance · Labs

Source: split from `med-tech-curriculum-V1.md` (V1). From this split onward this chapter is maintained **independently** here. See `Volume-III-Chapter-16-context.md` for the AI interaction log.

Prerequisites: Ch 12, Ch 15, Ch 3.

Learning outcomes — the student can: deploy models locally (Ollama/Open WebUI); build an inference API; integrate via FHIR (read/write, SMART-on-FHIR); monitor performance/drift; navigate the regulatory/compliance landscape.

Labs (hands-on) — 16 h:

#	Lab	h
16a	Deploy a model with Ollama + Open WebUI	3
16b	Wrap the model in a REST inference API (FastAPI)	4
16c	FHIR integration: read patient context, write results back	5
16d	Add monitoring/logging & a drift check	4

Datasets/tools: Ollama, Open WebUI, FastAPI, HAPI FHIR/Synthea; monitoring tools.

Assessment: deployed FHIR-integrated service (**60%**); monitoring setup (**20%**); quiz (**20%**).

Key decisions: local vs. cloud; API design; regulatory pathway; PHI handling at inference.

References: plan §13 → ...*local inference*; *Interoperability standards*; *Clinical data, code sets & privacy*. **Hours:** Theory **10** + Lab **16** = **26**.